

Ziyue Yang

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EDUCATION BACKGROUND

MSc Financial Technology, University College London

Sep 2024 – Dec 2025

Core Courses: Data-driven Modelling of Financial Market | Advanced Machine Learning in Finance | Database Fundamentals | Algorithmic Trading | Innovation and Strategy in Finance | Blockchain Technologies

BSc Financial Mathematics, First Class Honours, University of Liverpool

Sep 2020 – Jul 2024

Core Courses: Statistical Distribution Theory | Risk Management | Financial Engineering | Linear Statistical Models | Numerical Analysis | Micro & Macroeconomics | Stochastic Process and Calculus

WORK EXPERIENCES

Credit Risk Modelling Intern | Samoyed Financial, Shanghai, China

Jul 2025 – Oct 2025

- Developed a LightGBM-based binary classification model to identify customers eligible for 24% interest-rate products; achieved strong performance with out-of-time (OOT) KS of approximately 0.33 after applying Bayesian Optimization for hyperparameter tuning.
- Built a long-delinquent risk identification model using XGBoost; achieved OOT KS of 0.24, with stratified backtesting confirming higher stability and discriminatory power across different customer segments.
- Engineered about 500 new modeling features derived from credit reports; developed modular Python functions (e.g., time-series computation, result encapsulation, fallback logic) to automate feature generation workflows, improving feature-construction efficiency by about 40% and enhancing development stability and security.

ESG Research Intern | Development International, London, UK

May 2025 – Jul 2025

- Collected and cleaned ESG and financial data for STOXX Europe 600 constituents ranging from 2014 to 2024 via the Refinitiv Python API, constructing a final panel dataset of 4,835 firm-year observations.
- Built and benchmarked multiple models, including GLM, Decision Tree, Random Forest, GBDT, and XGBoost with hyperparameter tuning via cross-validation and random search; the XGBoost model achieved the best out-of-sample performance ($R^2 \approx 0.68$, $RMSE \approx 2.9$), demonstrating strong predictive power for corporate profitability.
- Applied SHAP, Permutation Importance, PDP, ALE, and ICE techniques to interpret model behavior, revealing nonlinear positive threshold effects of environmental and social dimensions at higher ESG scores; identified industry heterogeneity and the critical role of key financial indicators in shaping profitability forecasts.

Equity Research Intern | Sealand Securities Research Institute, Shanghai, China

Feb 2023 – Jun 2023

- Created a 53-page research report on a detailed analysis of a listed dairy giant "Yili" with organizational structure, core product competitiveness, and the macro retail environment based on its 2023 annual report and prospectus.
- Developed comprehensive dairy database exceeding 100MB in Microsoft Excel, including key metrics such as market size, retail volume, and gross profit margin. The organized data was utilized in 7 research reports.

Operations & Risk Analysis Intern | Agricultural Bank of China, Zhejiang, China

Jun 2022 – Aug 2022

- Supported risk assessment and credit analysis for personal and auto loans by validating financial documents and compiling key risk variables (loan amount, term, interest rate, LTV ratio), resulting in a 40% improvement in loan processing efficiency.
- Contributed to investment product marketing by interpreting client feedback and tailoring product explanations based on risk-return profiles.

PROJECT EXPERIENCES

Leveraged Algorithmic Trading Strategies on Treasury ETFs

Mar 2025 – Apr 2025

- Designed and backtested two leveraged algorithmic trading strategies: Dual Moving Average Crossover and Bollinger Bands, based on excess returns of the SPTL ETF over the effective federal funds rate.
- Constructed daily trading signals using Python by applying rolling windows for moving averages and volatility bands; found that the trend-following DMA strategy outperformed with a Sharpe Ratio of 0.97 and Calmar Ratio of 2.17.

CIR–HJM Interest Rate Modelling and Simulation

Oct 2023 – Dec 2023

- Developed and calibrated the Cox–Ingersoll–Ross (CIR) short-rate model using historical government bond yield data, estimating parameters via regression and Excel Solver to fit yield curves across maturities, and extended it into the Heath–Jarrow–Morton (HJM) framework to model yield curve and forward rate dynamics.
- Implemented Monte Carlo simulations in Python using variance reduction techniques such as Control and Antithetic Variates with 200 time steps, verified mean reversion and drift–diffusion consistency within 3% and 1% analytical deviations, and improved estimator stability by over 35%.

SKILLS / INTERESTS

Quantitative & Technical Skills: Python (Proficient) | SQL (Proficient) | Power BI (Proficient) | R (Intermediate)

Languages: English (C1 – Advanced, CEFR) | Mandarin (Native) | Spanish (Basic)

Interests: Photography | Tennis | Yoga | Badminton | Swimming